Monday - take-home for Test 2 is due

No ~~office~~ student hours ~~next~~ Thurs.

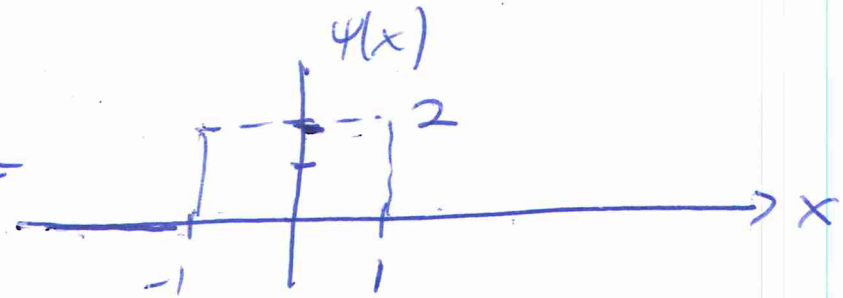
Go to the MLC & work with class mates!

Read the text book

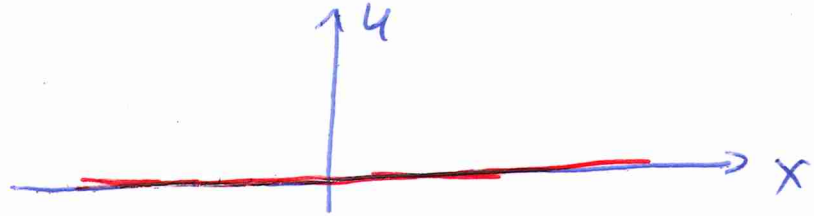
Next Friday Jon will be guest lecturing.

Graphs of sol'ns

ex) $\phi(x)=0$ $\quad \psi(x) =$



$t=0$

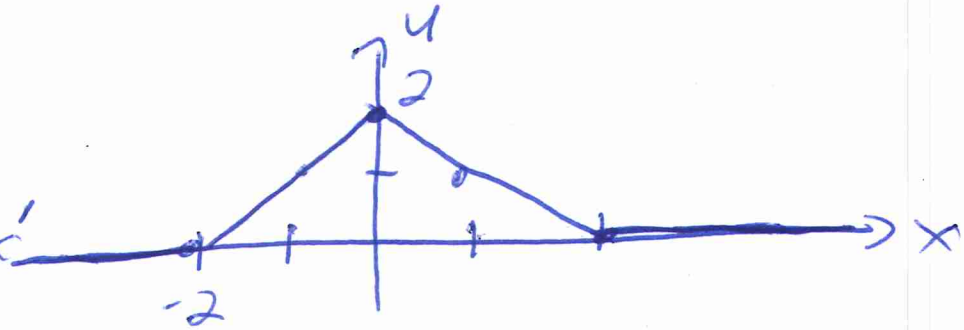


$t=0$

$$u(x,t) = \frac{1}{2}\int_{x-t}^{x+t} \psi(x')\,dx'$$

$t=1$

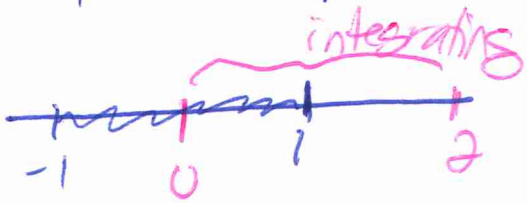
$$u(x,1) = \frac{1}{2}\int_{x-1}^{x+1} \psi(x')\,dx'$$



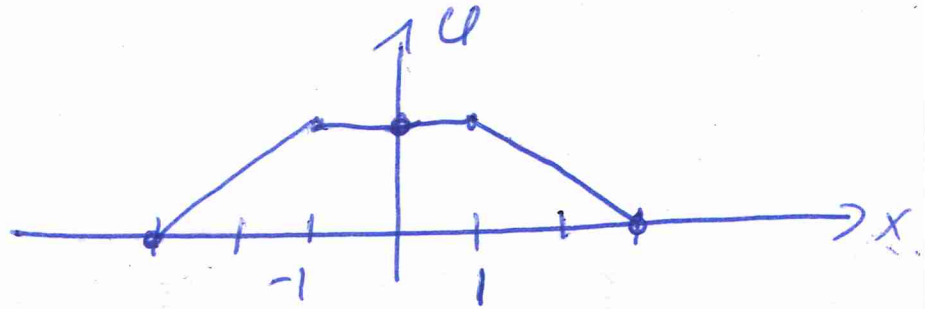
$t=1$

$x>2$

$x=1$



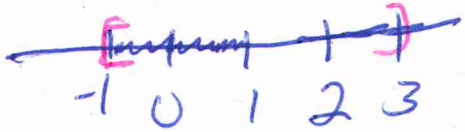
$$u(x,2) = \frac{1}{2}\int_{x-2}^{x+2} \psi(x')\,dx'$$



$t=2$

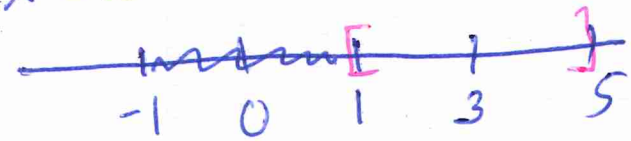
$x=0 \quad u=2$

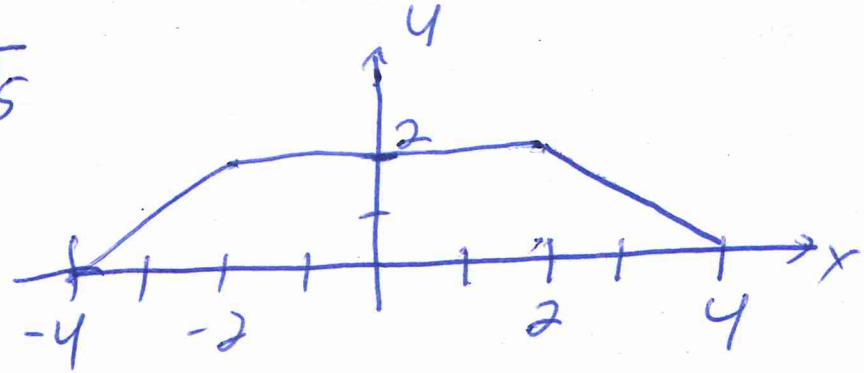
$x=1$



$\int_{-1}^{1}$

$x=3$





$t=3$

~~ex) $\phi_{\xi} = \psi = 0$ when~~

characteristics: $\xi_x^2 - 1^2 = 0 \qquad \xi_x = \pm 1$

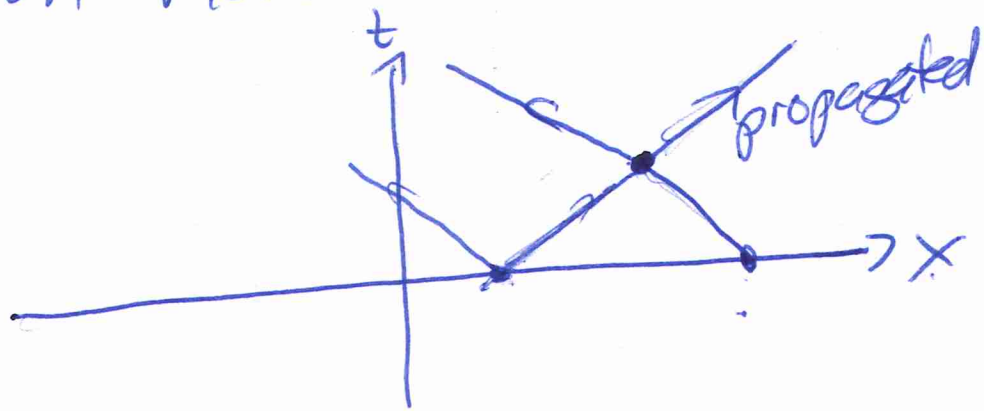
$\underline{\xi} = (\pm 1, 1)$

"surface" has $\underline{\xi}$ as its normal?

line that is $\perp$ to $(1,1)$, $(-1,1)$

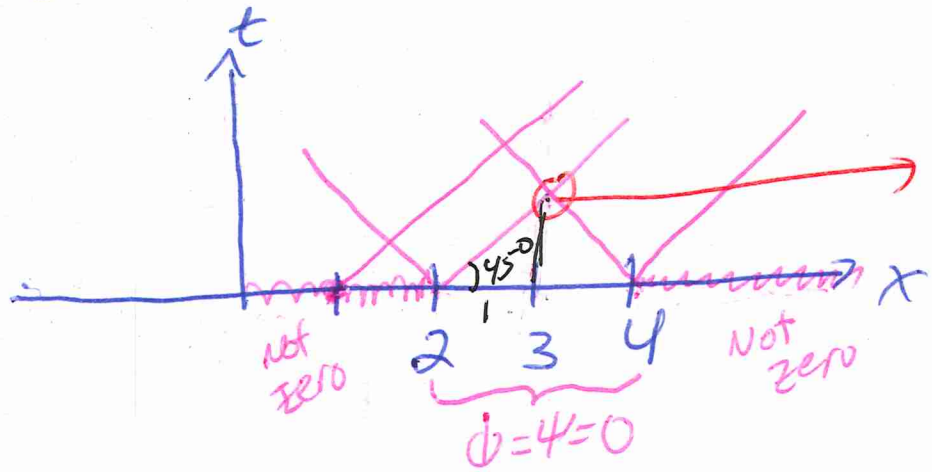
$(1,1)$: $c(-1,1) \qquad (-1,1)$: $c(1,1)$

information moves in the lines $\parallel$ $\langle 1,1 \rangle$ & $\langle -1,1 \rangle$



ex) $\phi = \psi = 0 \qquad 2 < x < 4$

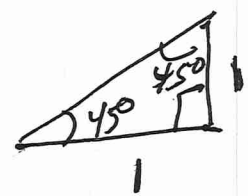
How long are we guaranteed that $u(3,t) = 0$



$u(3,0) = 0$

Find this time!

$t = 1$



Ans: $t < 1$

$$u(3,t) = \frac{1}{2}\left[\phi(3+t) + \phi(3-t)\right] + \frac{1}{2}\int_{3-t}^{3+t} \psi(x')\,dx'$$

$$u(3,1) = \frac{1}{2}\left[\underbrace{\phi(4)}_{\text{could be } \neq 0} + \underbrace{\phi(2)}_{\text{"}}\right] + \frac{1}{2}\int_{2}^{4} \underbrace{\psi(x')}_{0}\,dx'$$

ex) $u_{xx} - c^2 u_{tt} = 0 \Rightarrow u_{xx} - u_{t't'} = 0$

$t' = ct \qquad \frac{\partial u}{\partial t} = \frac{\partial u}{\partial t'}\frac{dt'}{dt} = c\,u_{t'}$

$t' = \frac{1}{c}t \qquad u_t = \frac{1}{c}u_{t'} \qquad u_{tt} = \frac{1}{c^2}u_{t't'}$

$u(x,0) = \phi(x)$

$u_t(x,0) = \psi(x)$

$\frac{1}{c}u_{t'}(x,0) = \psi(x)$

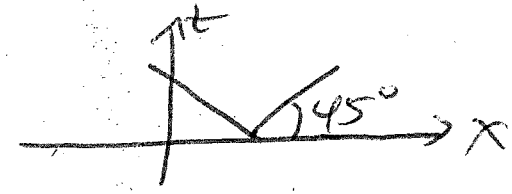
$u(x,0) = \phi(x)$

$u_{t'}(x,0) = c\psi(x)$

$$u(x,t') = \frac{1}{2}\left[\phi(x+t') + \phi(x-t')\right] + \frac{1}{2}\int_{x-t'}^{x+t'} c\psi(x')\,dx'$$

$t' = \frac{1}{c}t$

§8.3 Energy method

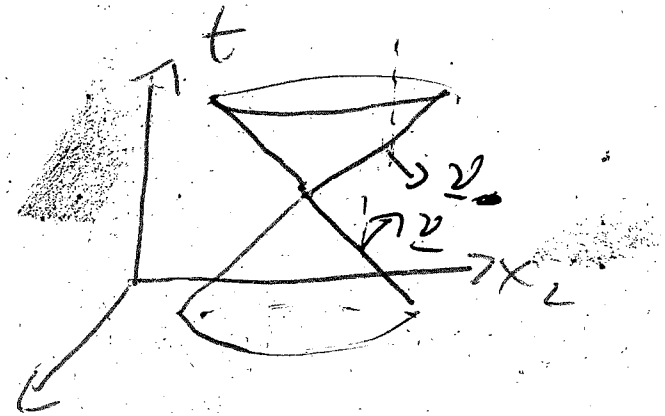


what is the analogous of in higher dimensions?

characteristic directions $\nu_t = 1$ $\quad \underline{\nu}$

$$\nu_1^2 + \dots + \nu_n^2 - \nu_t^2 = 0 \Rightarrow \nu_1^2 + \dots + \nu_n^2 = 1$$

This means $\underline{\nu}$ makes a 45° angle with the t-axis.



why? $\underline{e}_t = \langle 0, 0, 1 \rangle$

$\underline{\nu} = \langle \nu_1, \nu_2, 1 \rangle$

Recall $\underline{u} \cdot \underline{v} = |\underline{u}|\,|\underline{v}| \cos\theta$

$$\Rightarrow \underline{e}_t \cdot \underline{\nu} = |\underline{e}_t|\,|\underline{\nu}| \cos\theta$$

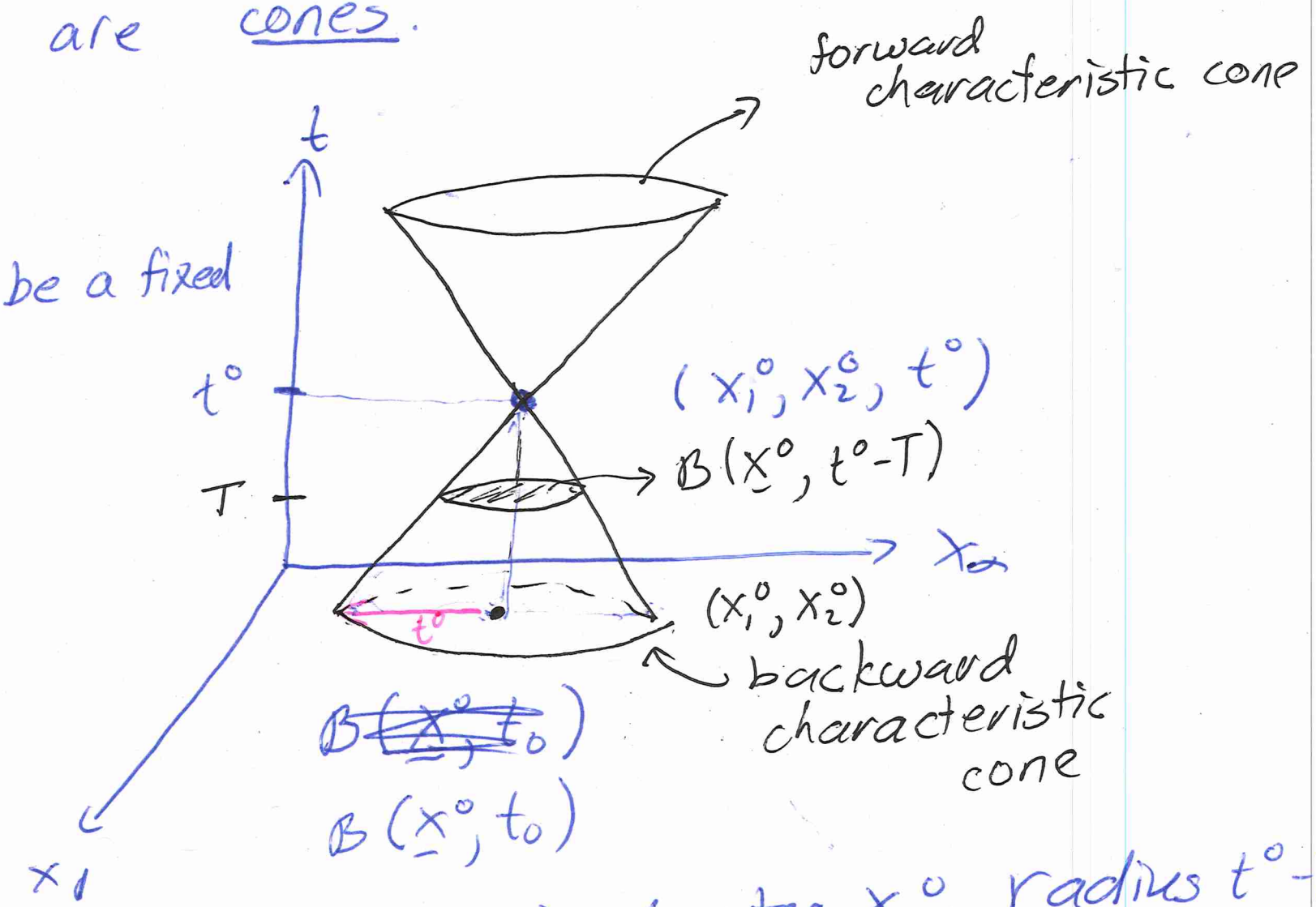
$$1 = (1)(\sqrt{\underbrace{\nu_1^2 + \nu_2^2}_{1} + 1})\cos\theta$$

$$1 = \sqrt{2}\cos\theta, \quad \cos\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$

Characteristic surfaces that are $\perp$ to $\underline{\nu}$ are <u>cones</u>.

Let T be a fixed time.



Define $\overline{B}(\underline{x}^0, t^0 - T)$ (center $\underline{x}^0$, radius $t^0 - T$) be the intersection of the plane $t = T$ and the backward cone.

<u>Thm 3.1</u> Suppose $u \in C^2(\bar{\Omega})$ satisfies the wave eqn. Then

$$\int_{B(\underline{x}^0, t^0 - T)} \left(u_{x_1}^2 + u_{x_2}^2 + \dots u_{x_n}^2 + u_t^2 \right) \Big|_{t=T} dx$$

$$\leq \int_{B(\underline{x}^0, t^0)} \left(u_{x_1}^2 + \dots + u_{x_n}^2 + u_t^2 \right) \Big|_{t=0} dx$$